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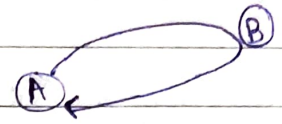
TCP Timer Management

new expected time → expected ~~time~~ RTT

$$SRTT_{t+1} = \alpha \, SRTT_{t} + (1-\alpha) \, R$$

SRTT: Smooth RTT; α: Smoothing factor; R: ~~ins~~ instantaneous (observed) RTT



Avg time = 60 min

* unknown - congestion / despite knowing distance and speed

→ today ~~ds~~ = 80 mins

~~→ the path~~

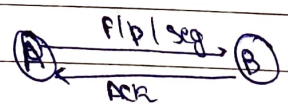
* Smoothin factor ($0 \le \alpha \le 1$) which is more important
(typically we assume $\alpha = 7/8$ if not given)

RTTVAR (expected variance)

* not do anything till the expected variance, after time elapsed which you take some action

$$RTTVAR = \beta \, RTTVAR + (1-\beta) \, |SRTT - R|$$

β = is typically 3/4 $\quad 0 \le \beta \le 1$



RTO [Retransmission Time Out] = SRTT + 4 RTTVAR

* Timer will time out and you will do retransmission after this time.

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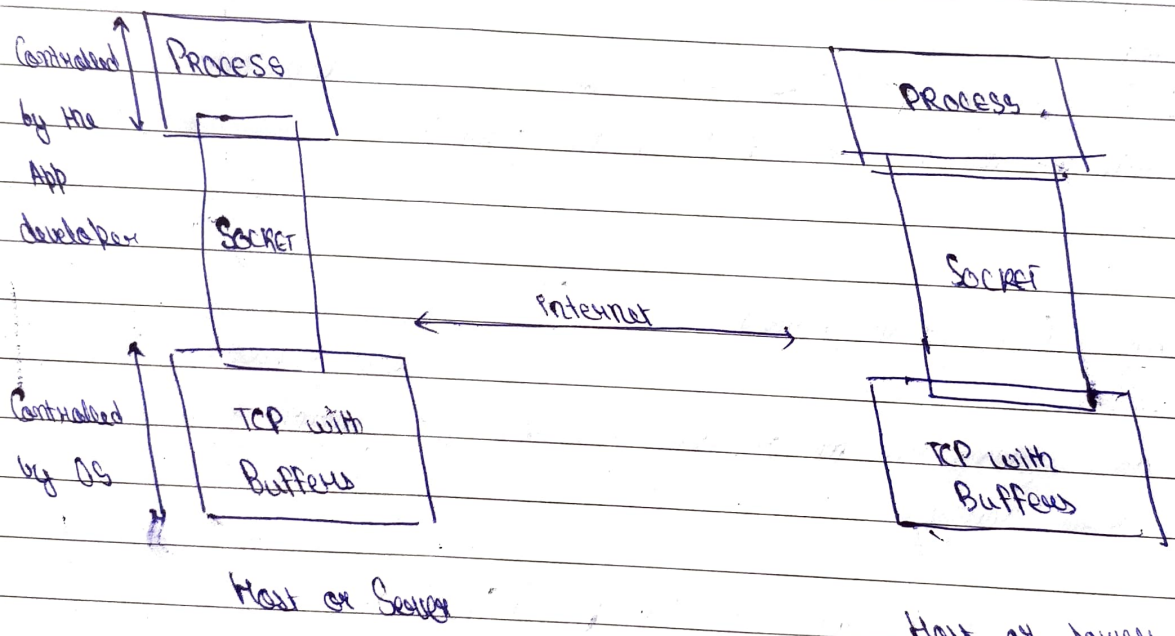
Application layer :

1. Client Server Model :

Client → webbrowser (firefox)

Server → webserver (apache)



Client Actions :

1. Creates a socket
2. Maps server name to IP address (get host by name)
3. Connect to a given port on the server address [connect]

4. Client initiates the communication with the server.

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Server Actions :

1. Creates a socket (Socket ())
2. Bind to one or more port numbers (Bind ())
3. Listen on the socket (listen ())
4. Accept client connections (Accept ())

* Server process must be Running

① Socket ()

② Connect ()

The connect function is used by TCP client to establish a connection with a TCP server. The function returns 0 if it succeeds in establishing a connection (successful three way handshake) or it returns -1 otherwise.

The client does not have to call bind function before calling the connect function. The kernel will choose both an ephemeral port and source IP if necessary.

③ Bind ()

The bind assigns a local protocol address to a socket. With the Internet protocols, the address is the combination of an IPV4 or an IPV6 address along with 16 bit TCP port number.

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Bind returns 0 if it succeeds and -1 on error.

④ Listen ()

→ The listen function call converts a socket into a passive socket indicating that the kernel should accept incoming connection request directed to this socket. Returns 0 if it succeeds and -1 on error.

⑤ Accept ()

→ In the call to accept function, the server is put to sleep and when for an incoming client request the three way handshake is complete, the function accept wakes up and returns the socket descriptor representing the client socket. The call to accept function is run in an infinite loop so that the server is always running and the delay or sleep of 1sec (approx) ensures that the server does not eat up all our CPU resources.

SERVER ARCHITECTURE :

1. Connection oriented : reliable but needs OS resources.
2. Connectionless : needs less resources but application has to handle loss of messages.

SERVER STATE :

1. STATELESS : Each transaction is independent
2. STATEFUL : Server maintains the state. This is faster but expensive for servers.

DELTA Notebook

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APPLICATION LAYER PROTOCOLS :

HTTP : Hyper text transfer protocol (Stateless protocol)
→ TCP : underlying protocol.
→ Port number : 80

Q. What is the difference b/w persistent and non persistent http connection ?

→ Non persistent - requires connection setup again and again for each objects to send. (HTTP 1.0)

→ Persistent does not require connection setup again and again. Multiple objects can use connection. (HTTP 1.1)

Q. What is pipelining in the context of HTTP ? Can pipeline be used in a non persistent HTTP ?

* HTTP pipelining is a technique in which multiple HTTP requests are sent on a single TCP connection without waiting for corresponding responses. No, pipelining cannot be used in non persistent HTTP.

1 COOKIES

2 WEB CACHING

3 FTP [port no : 20 & 21]
→ underlying protocol - TCP
→ Stateful protocol
→ two lines : ① Data ② Control

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4) SMTP (Simple Mail transfer protocol)

→ TCP

→ port no : 25

→ Sends from sender mail server to a receiver mail server

Q. Compare and contrast HTTP and SMTP.

Parameter	HTTP	SMTP
Port no.	80	25
State	Stateless	-
Types of TCP connection	Both (Persistent non persistent)	Persistent
Type of protocol	Pull	Push
Types of transfer	Transfer files b/w web server and client	Transfers mail via mail servers
No. of TCP connection	1	1
Type of band ~~for~~ transfer	In band	In band

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5) IMAP or POP 3
(Internet Mail access protocol) (post office protocol)

6) DNS (Domain Name System)